

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

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Section Id :	640653143413
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	25
Number of Questions to be attempted :	25
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653406345
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532197355 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SOFTWARE ENGINEERING

(COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406537134532.  Yes

6406537134533.  No

Sub-Section Number :

2

Sub-Section Id :

640653406346

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406532197356 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A software company is developing a payroll system for 500 companies.

The analyst prepares a fixed list of 25 questions and verbally asks exactly the same questions to HR managers from every company.

Which requirement gathering technique is being used?

Options :

6406537134534.  Unstructured interview

6406537134535.  Semi-structured interview

6406537134536.  Structured interview

6406537134537.  Observation

Question Number : 3 Question Id : 6406532197357 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An online banking application computes customer interest. Due to a mistake in the source code, the formula

```
interest = principal * 0.50
```

was implemented instead of

```
interest = principal * 0.05
```

The incorrect statement in the source code is an example of:

Options :

6406537134538. ✖ Error

6406537134539. ✔ Fault

6406537134540. ✖ Failure

6406537134541. ✖ Test Case

Question Number : 4 Question Id : 6406532197358 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A ride-sharing company has completed a sprint.

During a meeting, the team demonstrates the completed driver tracking feature to customers and receives feedback.

Which Scrum activity is being performed?

Options :

6406537134542. ✖ Sprint Planning

6406537134543. ✖ Daily Scrum

6406537134544. ✔ Sprint Review

6406537134545. ✖ Sprint Retrospective

Question Number : 5 Question Id : 6406532197359 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A team completed the following story points in the last four sprints:

Sprint	Story Points
1	18
2	22
3	20
4	24

What is the average velocity?

Options :

6406537134546. ✖ 20

6406537134547. ✔ 21

6406537134548. ✖ 22

6406537134549. ✖ 23

Question Number : 6 Question Id : 6406532197360 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An e-commerce company maintains the following modules:

- Inventory Management
- Payment Processing
- Recommendation Engine

Each module is assigned a single, well-defined objective.

All three modules directly access and modify the same global dictionary data structure that stores product information. Any change to the structure of this shared dictionary requires modifications in all three modules.

Which statement is true?

Options :

- 6406537134550. ✓ High cohesion and high coupling
- 6406537134551. ✗ High cohesion and low coupling
- 6406537134552. ✗ Low cohesion and high coupling
- 6406537134553. ✗ Low cohesion and low coupling

Question Number : 7 Question Id : 6406532197361 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A developer notices that a video editing application is slow.

The team wants to identify:

- functions consuming maximum CPU
- memory hot spots
- performance bottlenecks

Which of the following debugging techniques should be used?

Options :

- 6406537134554. ✗ Logging
- 6406537134555. ✓ Profiling Tool
- 6406537134556. ✗ Stepping in debugger
- 6406537134557. ✗ Dump and Diff

Question Number : 8 Question Id : 6406532197362 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A university portal must support:

- student registration

- fee payment
- grade viewing

A new class named `UniversityPortal` is introduced with a single method:

```
performAdmission()
```

The method internally coordinates the registration, payment, and grade management subsystems to complete the admission process.

Which of the following design patterns is followed in this case?

Options :

- 6406537134558. ✖ Factory
- 6406537134559. ✖ Adapter
- 6406537134560. ✔ Facade
- 6406537134561. ✖ Builder

Question Number : 9 Question Id : 6406532197363 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A company develops an Online Travel Booking System.

1. Travelers use the mobile app to search and book flights.
2. Travel agents inside the company update airline contracts and pricing.
3. A payment gateway processes customer transactions.
4. The airline reservation system exchanges seat availability data with the application.

Which option correctly identifies the client type for each case?

Options :

- 6406537134562. ✔ 1-External User, 2-Internal User, 3-Software Client, 4-Software Client
- 6406537134563. ✖ 1-Internal User, 2-External User, 3-Software Client, 4-Software Client
- 6406537134564. ✖ 1-External User, 2-Software Client, 3-Internal User, 4-Software Client
- 6406537134565. ✖ 1-Software Client, 2-Internal User, 3-External User, 4-Software Client

Question Number : 10 Question Id : 6406532197364 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A software company releases a new digital banking application to 100 selected customers. The customers use the application in real-world conditions and report usability issues.

Which testing phase is being performed?

Options :

- 6406537134566. ✖ Unit Testing
- 6406537134567. ✖ Integration Testing
- 6406537134568. ✖ Alpha Testing
- 6406537134569. ✔ Beta Testing

Question Number : 11 Question Id : 6406532197367 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following requirements for an online examination system:

Requirement 1: *Students shall be able to submit answers online.*

Requirement 2: *The system shall process and save submissions within 2 seconds.*

Which statement is true?

Options :

- 6406537134578. ✖ Both are functional requirements.
- 6406537134579. ✔ Requirement 1 is functional and Requirement 2 is non-functional.
- 6406537134580. ✖ Requirement 1 is non-functional and Requirement 2 is functional.
- 6406537134581. ✖ Both are non-functional requirements.

Question Number : 12 Question Id : 6406532197368 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the user story as follow:

As a customer,

I want to track my food delivery,

So that I know when my order will arrive.

The story is refined as:

"...the delivery status must refresh every 15 seconds."

Which SMART property is improved?

Options :

- 6406537134582. ✖ Specific
- 6406537134583. ✖ Relevant
- 6406537134584. ✔ Measurable
- 6406537134585. ✖ Achievable

Question Number : 13 Question Id : 6406532197369 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A healthcare portal allows doctors to prescribe medicines and patients to book appointments. However, new users frequently struggle to discover how appointments can be cancelled because there is no guidance or tutorial.

Which usability goal is most affected?

Options :

6406537134586. ✖ Efficiency

6406537134587. ✔ Learnability

6406537134588. ✖ Safety

6406537134589. ✖ Memorability

Question Number : 14 Question Id : 6406532197370 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A UX designer creates clickable screens using Figma. Users can navigate between screens, enter sample data, and simulate workflows.

What type of prototype is this?

Options :

6406537134590. ✖ Storyboard

6406537134591. ✖ Paper Prototype

6406537134592. ✔ Interactive Prototype

6406537134593. ✖ Digital Mock-up

Question Number : 15 Question Id : 6406532197371 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An e-commerce website allows customers to browse products and compare prices without requiring login.

Which design heuristic is supported?

Options :

6406537134594. ✔ Freedom

6406537134595. ✖ Recognition over Recall

6406537134596. ✖ Consistency

6406537134597. ✖ Show Status

Question Number : 16 Question Id : 6406532197372 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Module A directly accesses and modifies the internal variables of Module B.

What type of coupling exists?

Options :

6406537134598. ✖ Data Coupling

6406537134599. ✖ Control Coupling

6406537134600. ✖ Common Coupling

6406537134601. ✔ Content Coupling

Question Number : 17 Question Id : 6406532197374 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Python function:

```
def check(x):  
    if x > 0:  
        if x % 2 == 0:  
            return "A"  
        else:  
            return "B"  
    else:  
        return "C"
```

What is the cyclomatic complexity?

Options :

6406537134606. ✖ 2

6406537134607. ✔ 3

6406537134608. ✖ 4

6406537134609. ✖ 5

Question Number : 18 Question Id : 6406532197375 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A compiler processes source code through the modules -- lexical analysis, parsing, semantic analysis, optimization, and code generation. The output of one module is used as input to the next module.

Which architecture style best matches this system?

Options :

6406537134610. ✖ MVC

6406537134611. ✖ Client-Server

6406537134612. ✔ Pipe-and-Filter

6406537134613. ✖ Peer-to-Peer

Question Number : 19 Question Id : 6406532197376 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A marketplace supports three delivery modes:

- Physical Shipping
- Email Delivery
- Instant Download

A single interface `deliverable()` forces every implementation to provide:

```
packageItem()  
shipItem()  
emailItem()
```

even when many methods are irrelevant.

Which SOLID principle is violated?

Options :

6406537134614. ✖ Single Responsibility Principle

6406537134615. ✖ Open-Closed Principle

6406537134616. ✔ Interface Segregation Principle

6406537134617. ✖ Dependency Inversion Principle

Sub-Section Number :

3

Sub-Section Id :

640653406347

Question Shuffling Allowed :

Yes

Question Number : 20 Question Id : 6406532197365 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A large organization develops software using teams located in Chennai, Bangalore, and Delhi. Before implementation begins, interface specifications are finalized.

Which statements about interfaces are correct?

Options :

6406537134570. ✔ They specify permitted requests and outputs.

6406537134571. ✖ They reveal the complete implementation details.

6406537134572. ✓ They allow independent development of components.

6406537134573. ✓ Implementation can change without affecting dependent modules if the interface remains unchanged.

Question Number : 21 Question Id : 6406532197366 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A startup is developing an AI-based legal document analysis system. Since requirements are unclear, they build a prototype and show it to lawyers for feedback.

Which are advantages of using a prototype model?

Options :

6406537134574. ✓ Helps clarify user expectations.

6406537134575. ✓ Helps identify design issues early.

6406537134576. ✗ Guarantees zero defects in the final product.

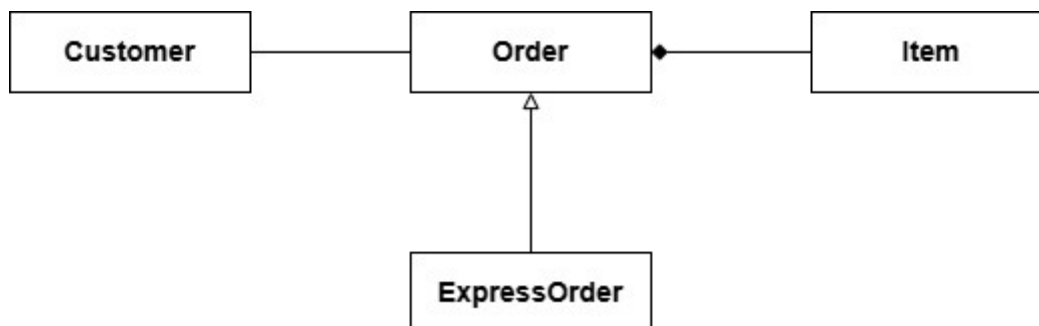
6406537134577. ✓ Enables early requirement refinement.

Question Number : 22 Question Id : 6406532197373 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the UML class diagram shown below.



Assume:

- Customer places Order s.
- Order contains multiple Item s.
- ExpressOrder is a special type of Order .

Identify the valid relationships.

Options :

6406537134602. ✓ Association between Customer and Order

6406537134603. ✓ Composition between Order and Item

6406537134604. ✓ Inheritance between **ExpressOrder** and **Order**

6406537134605. ✗ Composition between **Customer** and **Order**

Sub-Section Number : 4
Sub-Section Id : 640653406348
Question Shuffling Allowed : No

Question Id : 6406532197377 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (23 to 24)

Question Label : Comprehension

A company depends on a third-party fraud detection service for validating online transactions.

There is a risk that defects in the third-party service may impact the application.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 23 Question Id : 6406532197378 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

What type of risk is described?

Options :

6406537134618. ✗ Business Risk

6406537134619. ✓ Technical Risk

6406537134620. ✗ Project Risk

6406537134621. ✗ It is not a risk.

Question Number : 24 Question Id : 6406532197379 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which mitigation strategy is most suitable?

Options :

6406537134622. ✗ Conduct customer surveys

6406537134623. ✓ Benchmark and regularly inspect the external component

6406537134624. ✗ Explore the market

6406537134625. ✗ Reduce project scope

Question Id : 6406532197380 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (25 to 26)

Question Label : Comprehension

A company is developing an online furniture shopping platform. The requirements are well understood, the development team consists of highly experienced developers, and the estimated project size is 50 KLOC.

Project Type	a	b
Organic	2.4	1.05
Semi-detached	3.0	1.12
Embedded	3.6	1.20

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 25 Question Id : 6406532197381 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which COCOMO project category best describes the project?

Options :

- 6406537134626. ✓ Organic
- 6406537134627. ✗ Semi-detached
- 6406537134628. ✗ Embedded
- 6406537134629. ✗ Cannot be determined

Question Number : 26 Question Id : 6406532197382 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Using the Basic COCOMO model, what is the approximate initial effort for the project?

Options :

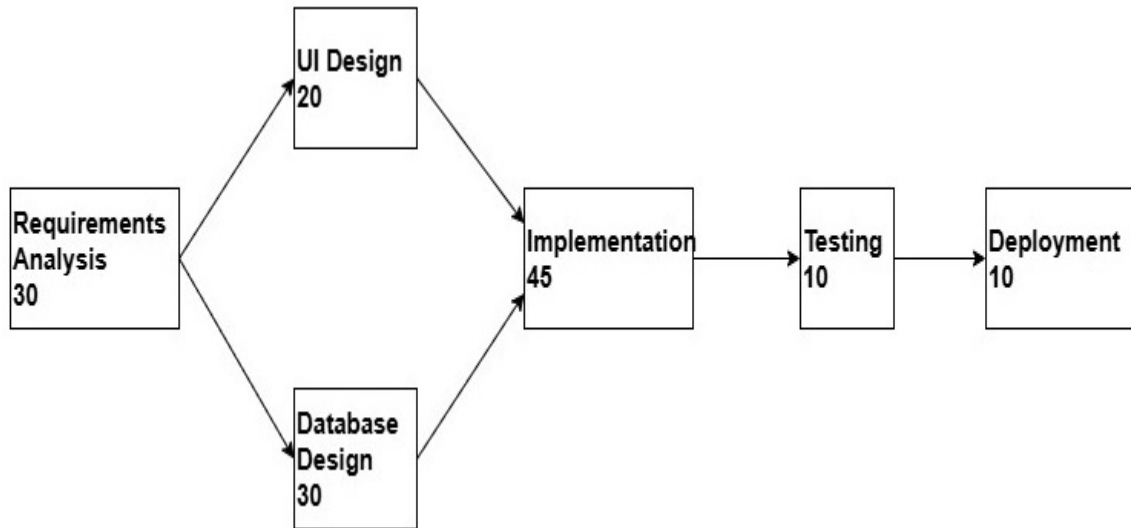
- 6406537134630. ✓ 146 Person-Months
- 6406537134631. ✗ 185 Person-Months
- 6406537134632. ✗ 220 Person-Months
- 6406537134633. ✗ 305 Person-Months

Sub-Section Number : 5
Sub-Section Id : 640653406349
Question Shuffling Allowed : No

Question Id : 6406532197383 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (27 to 28)

Question Label : Comprehension

A university is developing an online examination portal. Before implementation, the development team prepares the following activity network.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 27 Question Id : 6406532197384 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements are true?

Options :

- 6406537134634. ✓ *UI Design and Database Design can proceed simultaneously.*
- 6406537134635. ✗ *Testing can begin immediately after Requirements Analysis.*
- 6406537134636. ✓ *Implementation depends on both UI Design and Database Design.*
- 6406537134637. ✗ *Deployment can start before Testing is completed.*

Question Number : 28 Question Id : 6406532197385 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which task must be completed immediately before *Deployment*?

Options :

- 6406537134638. ✗ *UI Design*
- 6406537134639. ✗ *Database Design*
- 6406537134640. ✗ *Implementation*
- 6406537134641. ✓ *Testing*